

SmartFlex® Springs

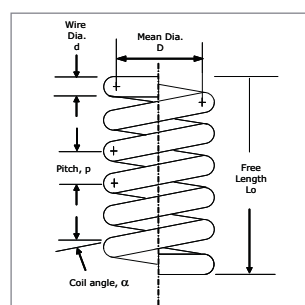
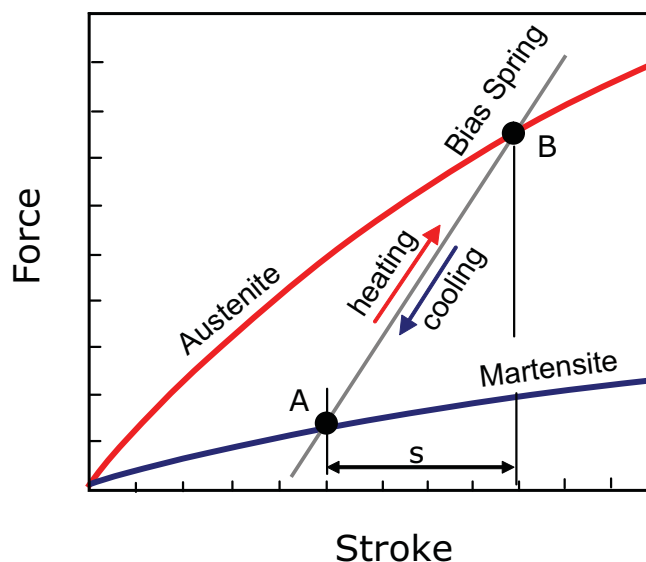
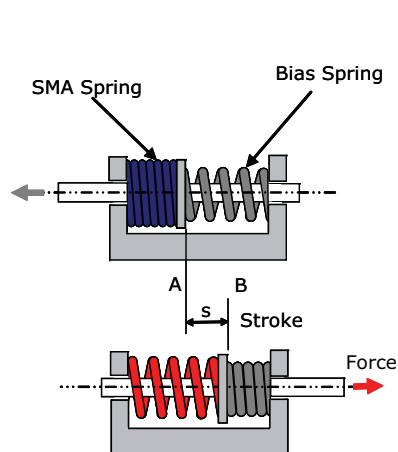
*"Shape Memory
Springs for Actuators"*

COMPRESSION SPRINGS



SMA compression springs can be used as very fast and miniaturized thermal-actuators. When heated above A_s (austenite start temperature) they expand developing an high force.

Typical behaviour of a SMA compression spring

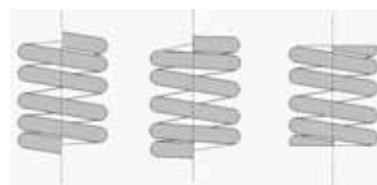


Mechanical and functional characteristics of some exemplifying compression spring

D [mm] Mean Diameter	L0 [mm] Free Length	d [mm] Wire Diameter	F [N] Typical Force	s [mm] Typical Stroke
14	130	1,5	20	30
10	25	1,5	14	8
9	16	2,2	50	2
6	18	1,5	30	3
3	8	0,6	2	2

SMA compression springs can be customized according to your design and application requirements.

Plain Ends **Closed Ends** **Closed Ends Ground**



Available Ends under request

SmartFlex® Springs

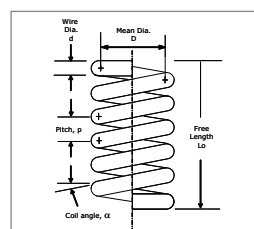
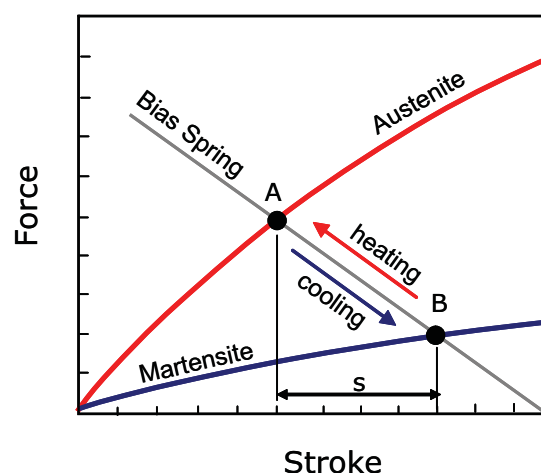
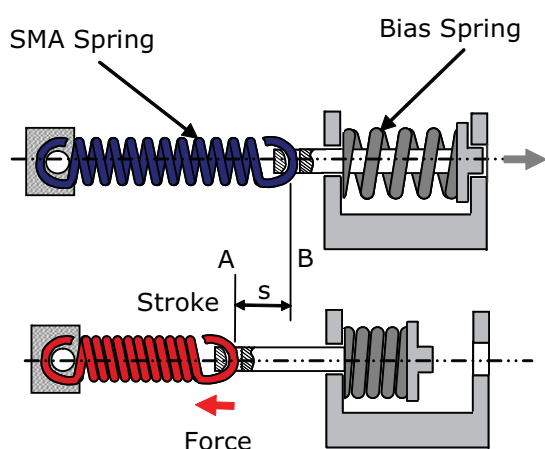
*"Shape Memory
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TENSILE SPRINGS



SMA tensile springs can be used as very fast and miniaturized thermal-actuators. When heated above A_s (austenite start temperature) they contract developing an high force.

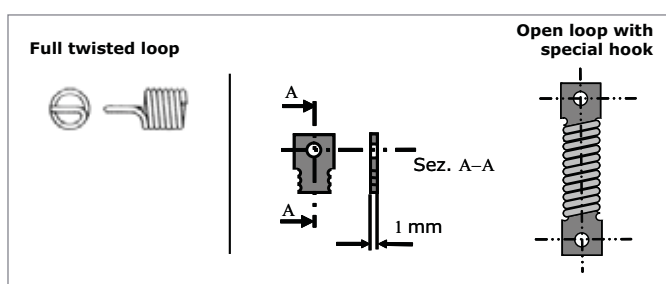
Typical behaviour of a SMA tensile spring



Mechanical and functional characteristics of some exemplifying tensile spring

D [mm] Mean Diameter	L0 [mm] Free Length	d [mm] Wire Diameter	F [N] Typical Force	s [mm] Typical Stroke
12	60	1,5	10	160
8	12	1,5	20	6
5	30	0,8	3	100

SMA tensile springs can be customized according to your design and application requirements.



Available Ends under request

The SAES Getters Group manufacturing companies are ISO9001 certified, the Asian and Italian companies are also ISO14001 certified.
Full information about our certifications for each company of the Group is available on our website at: www.saesgetters.com

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